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Arsalan Khan

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EDUCATION

University of Waterloo, *MASc in Electrical Engineering (Quantum Information)* **Sept. 2025 - Aug. 2027**
- **Cumulative GPA:** 3.88/4.00
- **Supervisors:** Thomas Jennewein, Michal Bajcsy

University of Toronto, *HBSc Physics Major, Computer Science Minor* **Sept. 2020 - May 2025**
- **Cumulative GPA:** 3.59/4.00

AWARDS & HONOURS

- **Undergraduate Research Fellowship**, University of Toronto – $5,000 CAD — May 2023
- **Natalia Krasnopolskaia Summer Undergraduate Research Fellowship** – $3,300 CAD *(declined)* — Apr. 2023
- **Victoria College Faculty Award** – $1,000 CAD *(Dean's List Scholar)* — Aug. 2022
- **Elizabeth (Eastlake) Vosburgh Scholarship** – $1,000 CAD *(Dean's List Scholar)* — Sept. 2021

RESEARCH & ENGINEERING EXPERIENCE

Graduate Researcher — Quantum Communication — Sept. 2025 — Present

Institute for Quantum Computing, University of Waterloo — *Supervisor: Dr. Thomas Jennewein*
- Developed from first principles and analytically validated a universal, reconfigurable architecture for heralding maximally entangled photonic qudits of arbitrary dimension in any discrete degree of freedom; created an algorithm that calculates the optical settings maximizing the heralding success probability.
- Developed Python software for automated polarization-state tomography of entangled photon pairs, controlling motorized waveplates, acquiring time-tagged coincidences, and reconstructing the source density matrix.

Quantum Optics Student — Sept. 2026 — Present

Canadian Space Agency
- Building a laboratory testbed and measurement workflow for characterizing single-photon emitters, beginning with quantum-dot and weak-coherent-pulse sources for satellite and fiber-based quantum key distribution (QKD).
- Supporting optical verification of the ground station for Canada's Quantum Encryption and Science Satellite (QEYSSat) mission.

Undergraduate Researcher — Quantum Photonics — May — Dec. 2023, Sept. — Nov. 2024

University of Toronto — *Supervisor: Dr. Amr S. Helmy*
- Investigated nonlinear waveguide sources for quantum photonics, including broadband difference-frequency generation and SPDC-based entangled-photon generation; contributed to a peer-reviewed publication and an undergraduate engineering thesis.
- Independently assembled and aligned free-space characterization setups using probe stations, thermal control, and optical spectrum analyzers; performed pump- and temperature-dependent measurements and diagnosed sources of experimental noise.
- Studied the generation and manipulation of time-energy-entangled photons and presented the work at the Canadian Undergraduate Physics Conference.

Optical Verification Engineer Intern — Jan. 2024 — Dec. 2024

Lumentum Operations LLC, *Design Verification and Testing*
- Planned and executed optical verification of wavelength-selective switches and multiplexers, characterizing polarization-dependent loss, spectral response, back-reflection, crosstalk, and insertion loss.
- Automated instrument control, data acquisition, and analysis by developing equipment drivers and test software in Python, C#, and VB.NET.
- Independently led a root-cause investigation into unexpected insertion-loss spectra, isolating contributions from air gaps, faulty laboratory fibers, wavelength-dependent bend loss, and coupling variability; the findings informed product-design and fiber-supplier decisions.

Undergraduate Researcher — Dark Matter + ML — Sept. 2023 — Apr. 2024

SuperCDMS Collaboration, University of Toronto — *Supervisor: Dr. Pekka Sinervo*
- Developed and evaluated TensorFlow models for simulated phonon measurements from silicon and germanium dark-matter detectors, identifying artifacts and limitations in the simulation pipeline.
- Reported the methods and findings in an undergraduate research thesis and oral conference presentation.

Research Assistant — Spectrometry for Lighting — May 2022 — Dec. 2022

University of Toronto — *Supervisor: Dr. Alstan Jakubiec*
- Designed, built, and calibrated a wearable optical spectrometer using C++, Python, and off-the-shelf components; benchmarked its performance for field studies of architectural lighting.

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PUBLICATIONS & RESEARCH OUTPUTS

Peer-Reviewed Journal Articles:

- B. Janjua, Z. M. Leger, J. Atkinson, M. L. Iu, **A. Khan**, J. H. Schmid, P. Cheben, and A. S. Helmy, "Difference Frequency Generation in Edge Emitting Photonic Crystal Lasers," J. Lightwave Technol., vol. 42, no. 17, pp. 5996–6002, 2024, doi: 10.1109/JLT.2024.3406253.

Conference Proceedings:

- J. Schrier, **A. Khan**, J. Ehling, P. Godin, T. Jennewein, "A technical overview of a weak coherent pulse source for satellite quantum key distribution with QEYSSat," Proc. SPIE, vol. 14143, Photonics for Quantum 2026, Art. no. 141430P, Jul. 22, 2026, doi: https://doi.org/10.1117/12.3112057.

Theses:

- **Undergraduate Thesis (ECE):** A. Khan, "Expanding Quantum Horizons: Monolithically Integrated C+L-Band Entangled Photon Source for Next-Gen Optical Networks", BSc thesis, The Edward S. Rogers Sr. Department of Electrical and Computer Engineering, University of Toronto, Dec 2023.
- **Undergraduate Thesis (Physics):** A. Khan, "Neural Networks for Improved Dark Matter Detection: Advances in SNOLAB SuperCDMS Detector Resolution", Undergraduate Research Project, Department of Physics, University of Toronto, Apr 2024.

Conference Presentations:

- Poster: J. Schrier, **A. Khan***, J. Ehling, P. Godin, T. Jennewein, "A Technical Overview of a Weak Coherent Pulse Source for Satellite Quantum Key Distribution with QEYSSat", at SPIE Photonics for Quantum 2026 (June 2026), and Quantum Days 2026 (February 2026).
- Talk: **A. Khan***, "Improving Dark Matter Detection with Deep Learning", at UofT Undergraduate Physics Conference (April 2024).
- Poster: Asad Khan*, **Arsalan Khan***, I. Lu*, "QuSparse: Quantum Approaches to Sparse Signal Reconstruction", at CUCAI 2024 (March 2024).
- Talk: **A. Khan***, Z. Leger, M. L. Iu, T. Stirling, A. Helmy, "Towards an integrated self-pumped source of hyperentangled photon pairs", at CUPC 2023 (October 2023).
- Talk: A. Jakubiec*, S. Wali*, **A. Khan***, "Development of an Open, Wearable Optical Spectral Irradiance Sensor" at the CORM CIE 2022 (November 2022).

*Presenting author.

LEADERSHIP AND ACADEMIC ENGAGEMENT

Conference Co-Chair — **Aug. 2026 — Present**

Canadian Graduate Quantum Conference 2027 (CGQC 2027)

- Co-leading a 10-member committee organizing a four-day national conference for approximately 100 graduate quantum researchers at the Institute for Quantum Computing and Perimeter Institute in May 2027.
- Coordinating fundraising, budgeting, venues, invited speakers, programming, communications, and cross-university outreach; the committee has secured $20,000 in funding to date.

Quantum Science Teaching and Outreach Volunteer — **Sept. 2025 — Present**

Institute for Quantum Computing, University of Waterloo

- Taught the Entanglement Lab at IQC's Undergraduate School on Experimental Quantum Information Processing (USEQIP), explaining Sagnac entangled-photon sources, projective measurements, and quantum-state tomography before supervising students through measurements and data analysis.
- Taught polarization concepts to children aged 12 and under through hands-on demonstrations for visiting students.

Quantum Machine Learning Developer — QuSparse — **Sept. 2023 — Apr. 2024**

University of Toronto Machine Intelligence Student Team (UTMIST)

- Implemented and evaluated quantum circuit Born machines using PennyLane, TensorFlow, and Python to investigate quantum-generative approaches to sparse signal reconstruction within a collaborative student research team.

Optics Team Member — FINCH CubeSat Mission — **May 2023 — Sept. 2023**

University of Toronto Aerospace Team (UTAT)

- Developed signal-to-noise and stray-light test plans for FINCH's hyperspectral imaging payload and coordinated data-storage and communications requirements with subsystem leads.

SELECTED TECHNICAL PROJECTS

Machine "Unlearning" (Dec. 2023) — Implemented and evaluated methods for removing selected training examples from image-classification models, including experiments with ensemble approaches on CIFAR-10 and MNIST.

Tunable FM Transmitter Circuit (Apr. 2023) — Designed, constructed, and tested a transistor-based FM transmitter incorporating an electret microphone, LC oscillator, and wire antenna.